Rounaq Choudhuri

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Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

DISSERTATION

Submitted in partial fulfillment of the requirements for the Degree:
M.Tech. in Artificial Intelligence & Machine Learning

By

Rounaq Choudhuri
(2023AA05218)

Under the supervision of
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(May, 2025)

Contents

1. List of Symbols & Abbreviation

Sl No.	Terminology	Abbreviation
1	Artificial Intelligence	AI
2	Machine Learning	ML
3	Deep Learning	DL
4	Generative Artificial Intelligence	GenAI
5	Variational Autoencoders	VAE
6	Generative Adversarial Networks	GAN
7	Autoencoders	AE
8	Human-Machine Interfaces	HMI
9	Skoltech Anomaly Benchmark	SKAB
10	Exploratory Data Analysis	EDA
11	Inter Quartile Range	IQR
12	Local Outlier Factor	LOF
13	k-Nearest Neighbors	kNN
14	True Positives	TP
15	False Positives	FP
16	False Negative	FN
17	True Negative	TN
18	Not a Number	NaN

Table 1: Terminology and Abbreviations

2. List of Tables

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Dissertation Title: Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

Name of Supervisor: Vishwanathan Raman

Name of Student: Rounaq Choudhuri

ID No. of Student: 2023AA05218

4. Abstract

Modern industrial environments, including manufacturing, oil and gas, and energy industries, are becoming increasingly complicated and automated, requiring extremely effective human-machine collaboration. Although existing Artificial Intelligence (AI) models have considerably improved human capabilities, attaining seamless synergy is hampered by intrinsic industrial data restrictions, specifically a lack of anomaly or failure data. Traditional Machine Learning (ML) and non-generative Deep Learning (DL) models, which are inherently discriminative, struggle to address these real-world data limitations because they cannot predict such circumstances. To address these limitations, this study analyzes the transformational potential of Generative Artificial Intelligence (GenAI) techniques other than Transformer based models.

Existing industrial processes often rely on predictive models trained on limited historical anomaly data, leading to suboptimal performance in rare event detection and a lack of comprehensive scenarios for operator training. To address this, the work first explores the critical role of Variational Autoencoders (VAEs) in robust anomaly detection within industrial data. Industrial systems frequently operate within a narrow range of normal parameters, making anomaly instances exceedingly rare. VAEs, as a generative model, are uniquely suited to learn the complex underlying distribution of normal operational states. VAEs can effectively identify deviations as anomalies, even when no or minimal actual anomaly data are available for training. This capability is pivotal for developing reliable early warning systems that improve the situational awareness of human operators and enable proactive decision-making in safety-critical contexts.

However, while abundant in normal operational states, real-world industrial data often lack the diverse and representative samples needed for comprehensive training and rigorous validation of advanced AI models and human operators. To address it, this work shifts its focus to synthetic data generation for training and validation, applying other GenAI methodologies, viz. Generative Adversarial Networks (GANs). The work leverages the framework for its specific strengths. GANs, through adversarial training, excel at producing highly realistic and diverse synthetic samples that mimic complex time-series data or rare

failure scenarios.

VAEs and GANs have the inherent potential to generate unique, coherent, and diverse data, in contrast to standard ML and non-generative DL techniques, which are discriminative and limited to learning from existing data. This generative capability is critical for overcoming the constraints of scarce or unbalanced industrial datasets, simulating unobserved scenarios, and providing an unparalleled means of robustly training AI systems and human personnel in difficult, data-poor industrial environments. This dissertation intends to demonstrate how these methods for generating and interpreting industrial data directly contribute to more confident decision-making in crucial scenarios by boosting human situational awareness. Consequently, enhancing human collaboration with reliable AI systems in complex industrial landscapes, thereby driving efficiency and safety in the era of Industry 4.0.

Key Words: Data Scarcity, Industrial Data, Anomaly Detection, Synthetic Data Generation, Generative AI

5. Literature Review

1. **Radicioni, L.; Bono, F.M.; Cinquemani, S. *"Vibration-Based Anomaly Detection in Industrial Machines: A Comparison of Autoencoders and Latent Space"*. Machines 2025, 13, 139.**
   This paper compares autoencoder-based methods for vibration-based anomaly detection in industrial machines, highlighting the limitations of reconstruction-based metrics for Variational Autoencoders (VAEs) and suggesting the need for latent space metrics.

2. **A. A. Pol, V. Berger, C. Germain, G. Cerminara and M. Pierini, *"Anomaly Detection with Conditional Variational Autoencoders"*, 2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA), Boca Raton, FL, USA, 2019, pp. 1651-1657.**
   This research introduces a method for conditional anomaly detection in production environments using Variational Autoencoders (VAEs). It focuses on providing interpretable *interesting scores* under real-time industrial constraints.

3. **Keskes, Mohamed. (2025). *"Generative Adversarial Networks for Synthetic Data Generation in Deep Learning Applications"*, Preprints, 2025. 10.20944/preprints202505.0836.v1.**

This review offers a comprehensive analysis of Generative Adversarial Networks (GANs) for synthetic data generation in deep learning applications. It explores their principles, advanced architectures like TimeGANs for time-series data, and their role in addressing data scarcity and privacy.

4. **H. Lu, M. Du, K. Qian, X. He and K. Wang, *"GAN-Based Data Augmentation Strategy for Sensor Anomaly Detection in Industrial Robots"* in IEEE Sensors Journal, vol. 22, no. 18, pp. 17464-17474, 15 Sept.15, 2022.** This paper proposes a GAN-based data augmentation strategy to improve sensor anomaly detection in industrial robots. It addresses the challenge of scarce failure data by generating synthetic samples for robust model training.

6. Detailed Plan of Work

Sl No.	Deliverables	Status	Start Week	End Week
1	Proposal Approval	Completed	1	1
2	Literature Review	Completed	1	1
3	Dataset Finalization	Completed	2	2
4	Data Wrangling	Completed	2	3
5	Data Sampling Preprocessing	Completed	3	3
6	Exploratory Data Analysis	Completed	3	3
7	Feature Engineering	Completed	4	5
8	Statistical Modelling	Completed	5	5
9	Statistical Model Evaluation	Completed	5	5
10	Machine Learning Modelling	Completed	6	6
11	Machine Learning Model Evaluation	Completed	6	6
12	Deep Learning Modelling	Completed	7	7
13	Deep Learning Model Evaluation	Completed	8	8
14	Generative Modelling	Ongoing	9	9
15	Generative Model Evaluation	Pending	10	10
16	Synthetic Data Generation	Pending	11	11
17	Synthetic Data Evaluation	Pending	12	12
18	Machine Learning Modelling with Synthetic Data	Pending	13	13
19	Machine Learning Model Evaluation	Pending	14	14
20	Comprehensive Evaluation Report	Pending	15	15
21	Dissertation Finalization	Pending	15	15
22	Submission	Pending	16	16

Table 4: Project Deliverables Schedule

7. Data Source

Introduction

Skoltech Anomaly Benchmark (SKAB) dataset is designed to evaluate the anomaly detection score. The SKAB dataset contains 35 data files in ".csv" format. Each file represents one experiment and contains one anomaly (the exception is the anomaly-free file, which does not contain any anomalies). The set of data represents a multidimensional time series collected from sensors installed on the test bench. This experiment was recorded 13 different types of anomalies across 3 days.

Column Description

- **datetime :** Dates and times of data points (YYYY-MM-DD hh:mm:ss).
- **Accelerometer1RMS :** Vibration acceleration along the x-axis.
- **Accelerometer2RMS :** Vibration acceleration along the y-axis.
- **Current :** Current strength on electric motors (Amperes).
- **Pressure :** Pressure in the circuit after the water pump (Bar).
- **Temperature :** Temperature of the engine housing ($°C$).
- **Thermocouple :** Temperature of the liquid in the circulation circuit ($°C$).
- **Voltage :** Voltage on electric motors (Volts).
- **RateRMS :** The circulation flow rate of the fluid inside the loop (Liter per minute).
- **anomaly :** Shows whether the point is anomalous (0 or 1).
- **changepoint :** Shows whether the point is the point of change of state (0 or 1).

Usage

Since the scope of this work is limited to anomaly detection, the changepoint column will be dropped. All the 35 data files will be merged into a single ".csv" file for the ease of processing and performing downstream tasks like Exploratory Data Analysis (EDA), Model Training and Evaluation, etc.

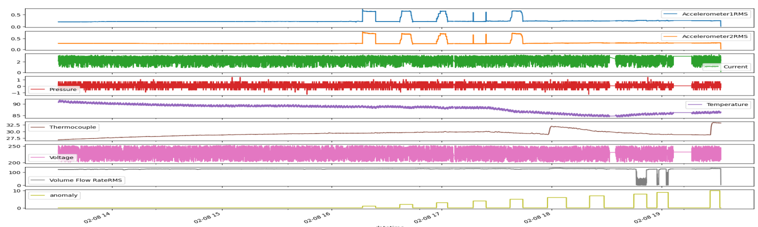
Source

Skoltech Anomaly Benchmark Dataset

datetime	Accelerometer1RMS	Accelerometer2RMS	Current	Pressure	Temperature	Thermocouple	Voltage	Volume Flow RateRMS	anomaly
2020-02-08 13:30:47	0.202394	0.275154	2.169750	0.382638	90.6454	26.8508	238.852	122.66400	0.0
2020-02-08 13:30:48	0.203153	0.277857	2.079990	-0.273216	90.7978	26.8639	227.943	122.33800	0.0
2020-02-08 13:30:49	0.203153	0.277857	2.079990	-0.273216	90.7978	26.8639	227.943	122.33800	0.0
...	...	...	...	...	...	...	...	...	...
2020-02-08 19:32:17	0.024764	0.037350	0.293163	0.382638	86.5109	33.2445	228.719	69.18940	0.0
2020-02-08 19:32:18	0.017131	0.016978	0.220920	0.054711	86.2620	33.2386	245.183	17.65750	0.0
2020-02-08 19:32:19	0.015752	0.015505	0.149842	0.382638	86.4799	33.2464	231.541	2.76765	0.0

(a)

Figure 1: Data collected across 3 dates



(a)

Figure 2: Time Series plot of the SKAB dataset

8. Data Preprocessing

8.1. Data Sampling

The SKAB dataset contains 8 sensor readings of a centrifugal pump spread across 3 days. These sensor readings are recorded multiple times in a second. Therefore, it was necessary that to sample the data into a small and manageable window of 1 second. After sampling the data into 1 second windows, the dataset now contains in total about 50,000 data points having 14 different types of data behavior including 13 anomaly types. Once, the data was sampled, the dataset was checked for “NaN” values and they were replaced by mean values usinf forward fill method.

8.2. Merging data

After sampling the dataset for 3 dates, these sub-datasets are merged into a single file containing 50,000 data points. The datasets before merger can be referenced in “Figure 1(a)”.

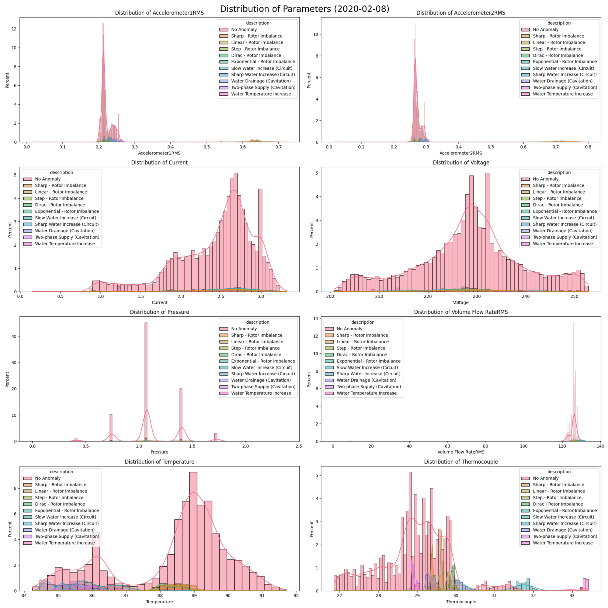


Figure 3: Histogram Plot for all parameters for every anomaly class (2020-02-08)

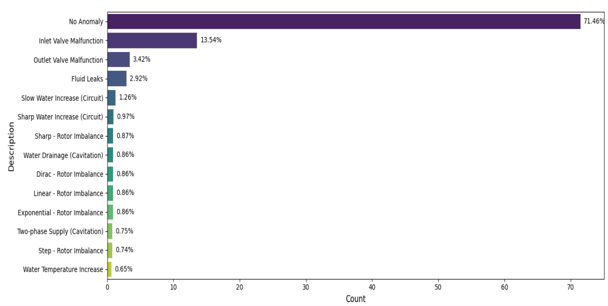


Figure 4: Proportion of Anomalies and Normal data point across all 3 dates

8.3. Time Series

The basic time series analysis can help to understand the behavior of the various sensors under different operational states and anomalous situations. The basic time series plots can be referenced in "Figure 2(a)".

9. Exploratory Data Analysis

9.1. Univariate Analysis

9.1.1 Histogram

"Figure 3" effectively visualizes the individual distributions of normal points and 13 distinct anomaly types across eight operational parameters. It highlights how different anomalies manifest as shifts or deviations from the normal operating ranges, offering insights into their

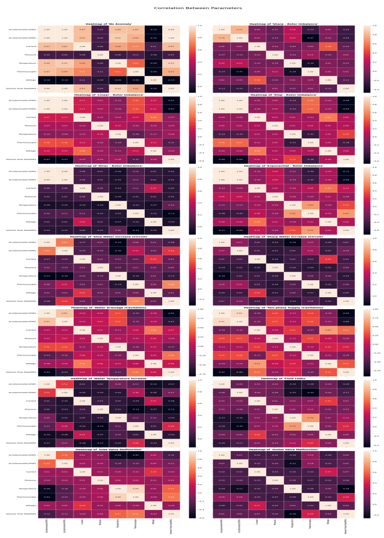


Figure 5: Correlation among the sensor parameters with respect to the different data point classes

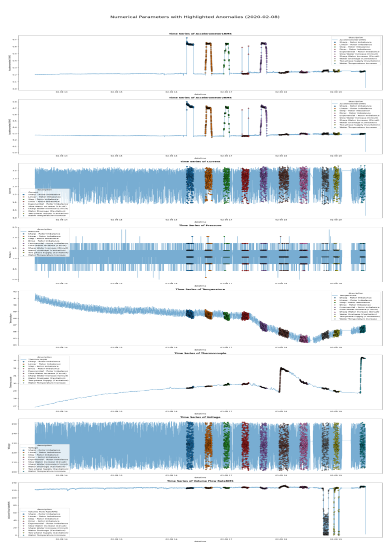


Figure 6: Time Series plot of 2020-02-08 highlighting various anomaly data points

unique signatures within each parameter.

9.1.2 Proportion of Data Points

"Figure 4" clearly illustrates a severe class imbalance within the dataset, where "No Anomaly" instances constitute a dominant 71.46%, while all ten anomaly types combined make up less than 30%. Such a highly imbalanced dataset, if used directly for training an anomaly detection model, will likely lead to a model biased towards the majority "No Anomaly" class, resulting in excellent performance on normal data but poor detection capabilities for rare anomaly types due to insufficient learning from their limited examples.

9.2. Bivariate Analysis

9.2.1 Distribution of Data Points by Anomaly Type

It can be inferred from that the different operational parameters behave significantly different in anomalous situations. This helps to identify the distribution and characteristics of the sensor parameters when operating at different operational states.

9.2.2 Correlation

The correlation heatmap in "Figure 5" clearly shows that while under normal operating conditions the sensor parameters have relatively high correlation. However, under anomalous situations the sensor parameters behave abnormally and almost no correlation with each other.

9.2.3 Time Series

"Figure 6"highlights the various anomaly points and its type for the 3 days of data. These figures gives a significant insight into the behavior of the pump under anomalous situations at various operational states. It can be observed that the anomaly points highly coincides within the data distribution range of normal data points, therefore, indicating an underlying issue with the system and making anomaly detection even more challenging.

9.3. Multivariate Analysis

9.3.1 Correlation among parameters by Anomaly Type

Similar to figure 3, "Figure 7" highlights that the distribution of anomaly data points coincides with the distribution of normal data points. The behavior of anomaly data points can be confirmed to be random or unexplainable using these parameters.

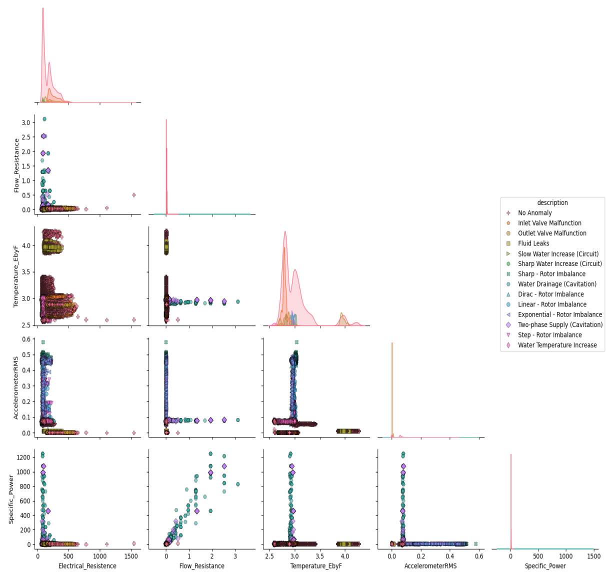


Figure 7: Correlation of basic interactive features of sensor parameters

10. Feature Engineering

10.1. Interaction Features

10.1.1 Electrical Resistance

- **What :** It is a measure of opposition to the flow of electric current in an electrical circuit, typically measured in ohms (Ω).
- **Why :** Deviations in calculated electrical resistance from expected values can indicate anomalies such as motor winding degradation, short circuits, or changes in mechanical load that affect current draw for a given voltage.

10.1.2 Flow Resistance

- **What :** It is a measure of the opposition to fluid movement through a system, analogous to electrical resistance, it represents the pressure drop required to achieve a certain flow rate.
- **Why :** Increase in flow resistance for a given flow rate or pressure can indicate blockages, increased friction due to scaling or debris, or issues with the pump itself, thus signaling an anomaly in the system.

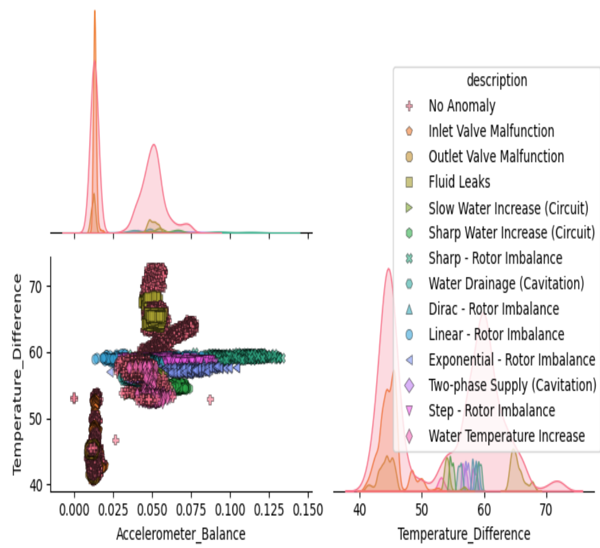


Figure 8: Correlation of differences of sensor parameters

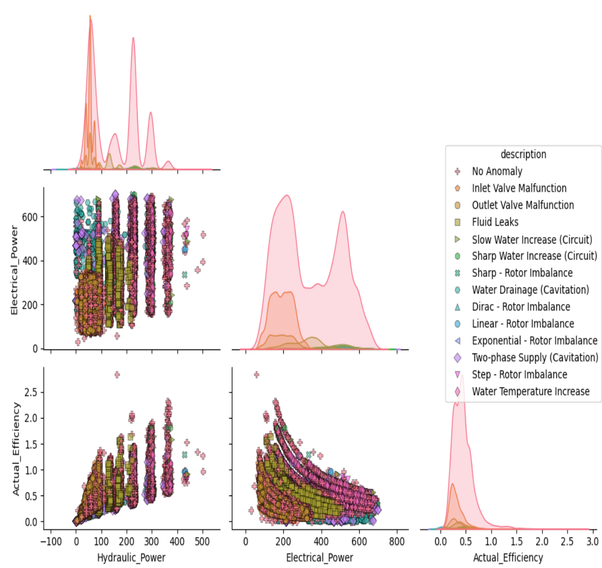


Figure 9: Correlation of efficiency and performance metrics

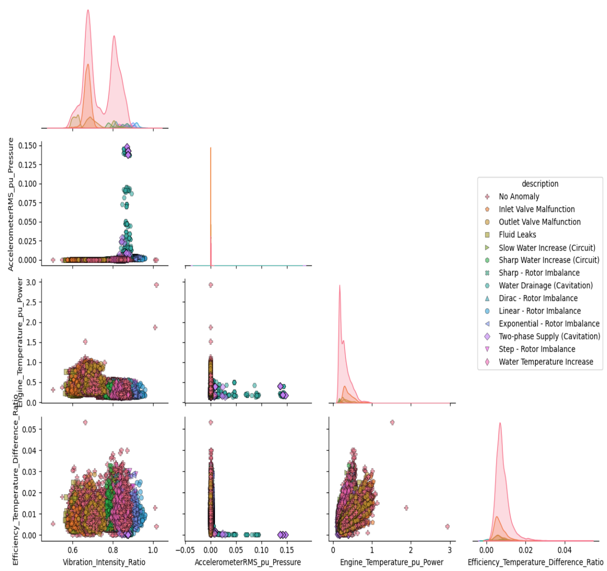


Figure 10: Correlation of higher order interactive feature of the sensor parameters

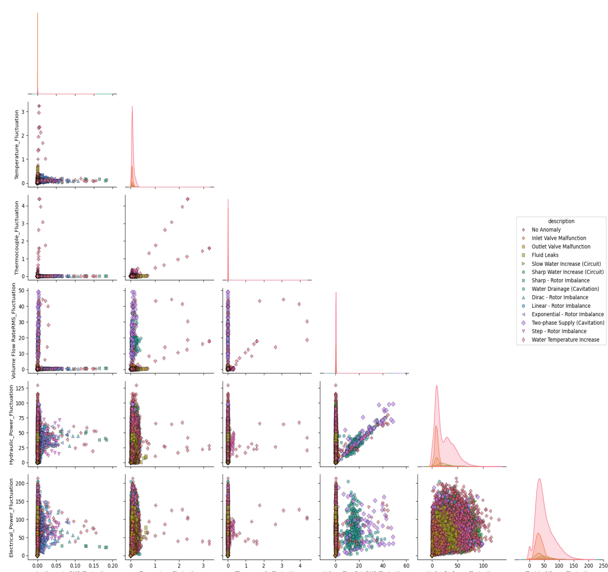


Figure 11: Correlation of fluctuation readings of the sensor parameters

10.1.3 Operational Temperature Ratio

- **What :** It represents the relationship between the temperature of the pump's engine body and the temperature of the fluid being circulated.
- **Why :** Significant deviations in this metric from a normal range can indicate issues such as inefficient heat transfer, motor overheating, or abnormal fluid heating due to cavitation, thereby signaling an anomaly.

10.1.4 Operational Vibration

- **What :** It represents the combined magnitude of vibration in two perpendicular directions on the pump, typically measured in g^2.
- **Why :** Increase in Operational Vibration beyond normal thresholds can directly indicate mechanical issues such as imbalance, misalignment, bearing wear, or cavitation within the pump or motor, thus serving as a strong anomaly indicator.

10.1.5 Specific Power

- **What :** It is a measure of the electrical power consumed by the pump per unit of fluid flow rate, indicating the energy required to move a certain volume of fluid, typically measured in Watt per Liter per Minute (W/(L/min)).
- **Why :** Increase in Specific Power for a given operating point can indicate reduced pump efficiency due to issues like wear, cavitation, or blockages, thereby signaling an anomaly.

10.2. Difference Features

10.2.1 Vibrational Balance

- **What :** It measures the disparity in vibration magnitudes between two perpendicular axes (x and y) on the pump, typically measured in g.
- **Why :** Increase in Vibrational Balance can indicate an imbalance in the system's vibration, potentially pointing to issues like misalignment or uneven bearing wear.

10.2.2 Temperature Balance

- **What :** It is the difference between the engine body temperature and the fluid temperature, indicating the thermal equilibrium or disparity within the pump system, typically measured in $°C$.

- **Why :** Change in Temperature Balance from its normal operating range can signal issues like inefficient cooling of the motor, excessive heat generation from the fluid (e.g., due to cavitation), or an external heat imbalance, thereby indicating an anomaly.

10.3. Efficiency & Performance Features

10.3.1 Hydraulic Power

- **What :** It is the measure of useful power transferred by the pump to the fluid, calculated as the product of fluid pressure, flow rate, and a conversion factor, representing the energy imparted to move the fluid, typically measured in kilo-Watt (kW).
- **Why :** Drop in Hydraulic Power relative to the electrical power input indicates a loss of pump efficiency or internal issues, serving as a strong indicator of an anomaly.

10.3.2 Electrical Power

- **What :** It is the measure of the rate at which electrical energy is transferred to the motor, calculated as the product of Voltage and Current (and power factor for AC systems), typically measured in kilo-Watt (kW).
- **Why :** Unexpected increases or fluctuations in Electrical Power, without corresponding changes in pump output, can indicate issues like motor overloading, winding problems, or increased mechanical resistance in the pump, thus signaling an anomaly.

10.3.3 Pump Efficiency

- **What :** It is the ratio of the hydraulic power delivered to the fluid to the electrical power consumed by the motor, indicating how effectively the pump converts electrical energy into useful fluid energy.
- **Why :** Decrease in Pump Efficiency for a given operating condition directly signals degradation, internal wear, cavitation, or blockages within the pump, making it a highly effective anomaly detector.

10.4. Higher Order Interactive Features

10.4.1 Vibration Intensity Ratio

- **What :** It is the ratio of vibration magnitudes from two different accelerometer sensors, indicating the relative vibration levels across different points of the pump.
- **Why :** Deviation of the Vibration Intensity Ratio from its expected value (often near 1 for balanced systems) can indicate an imbalance, misalignment, or a localized fault, thereby signaling an anomaly.

10.4.2 Vibration per Unit Flow

- **What :** It is a metric that relates the level of pump vibration to the volume of fluid being moved, showing vibration intensity relative to operational output.
- **Why :** Increase in Vibration per Unit Flow can occur when the flow rate is not proportionally high, it can point to issues like cavitation, internal impeller damage, or bearing problems not directly related to the hydraulic load, thus signaling an anomaly.

10.4.3 Engine Temperature per Unit Power

- **What :** It is a metric indicating the motor's operating temperature in relation to the electrical power it consumes, typically measured in $^{\circ}C$ / W.
- **Why :** Unusual increase in Engine Temperature Per Unit Electrical Power suggests issues such as motor winding problems, cooling system failure, or increased internal friction, thereby signaling an anomaly.

10.4.4 Efficiency & Temperature Balance Ratio

- **What :** It is a comparative metric that evaluates the relationship between the pump's efficiency and the difference between engine and fluid temperatures.
- **Why :** A disproportionate relationship between a drop in efficiency and the motor/fluid temperature balance can help distinguish between hydraulic problems (e.g., worn impeller) and electrical/motor-related issues, thus aiding in anomaly detection.

10.5. Fluctuation Features

10.5.1 Operational Vibration Fluctuation

- **What :** It represents the variability or instability in the product of the two accelerometer readings, indicating changes in the overall vibration pattern over time, typically measured in g^2.
- **Why :** Sudden or sustained increases in Operational Vibration Fluctuations can signal dynamic instabilities, loosening components, or the onset of faults like cavitation or bearing damage, thereby indicating an anomaly.

10.5.2 Temperature Fluctuation

- **What :** It represents the variability or instability in the temperature readings of the electric motor body.

- **Why :** Increased Engine Temperature Fluctuations can indicate inconsistent cooling, intermittent overloading, or electrical issues within the motor, thereby signaling an anomaly.

10.5.3 Thermocouple Fluctuation

- **What :** It represents the variability or instability in the temperature readings of the fluid within the circulation loop.
- **Why :** Elevated Fluid Temperature Fluctuations can suggest erratic flow, intermittent cavitation, or issues with external heat exchange in the fluid loop, thereby signaling an anomaly.

10.5.4 Volume Flow Rate Fluctuation

- **What :** It represents the variability or instability in the rate at which fluid is flowing through the circulation loop.
- **Why :** Increase in Volume Flow Rate Fluctuation can indicate cavitation, blockages, air entrainment, or issues with the pump's prime, thereby signaling an anomaly.

10.5.5 Hydraulic Power Fluctuation

- **What :** It represents the variability or instability in the useful power delivered by the pump to the fluid.
- **Why :** Increase in Hydraulic Power Fluctuation can point to intermittent issues with pump performance, such as partial cavitation, varying system resistance, or inconsistent fluid delivery, thereby signaling an anomaly.

10.5.6 Electrical Power Fluctuation

- **What :** It represents the variability or instability in the power consumed by the electric motor driving the pump.
- **Why :** Increase in Electrical Power Fluctuation can indicate unstable motor operation, intermittent electrical faults, or varying mechanical loads on the pump, thereby signaling an anomaly.

10.6. Correlation among Engineered Features

After engineering features relevant to the centrifugal pump domain it can be observed that some anomalies can be detected in higher-order engineered features. Deviation in distribution of anomaly data types is more prominent post feature engineering. However, most of the engineered features do not show high correlation, prompting the need for more complex engineered features or latent features through deep learning techniques. The behaviors of the anomalous points with respect to the engineered features can be reference in "Figure 8, 9, 10, & 11"

11. Modelling

11.1. Statistical Methods

11.1.1 Parametric

- **Z-Score Method**
  - **Description :** Measures how many standard deviations an observation is from the mean.
  - **Usefulness for Anomaly Detection :** Identifies anomalies as data points that fall beyond a certain number of standard deviations from the mean in a univariate distribution.
- **Multivariate Gaussian Model**
  - **Description :** Models data points as following a multi-dimensional Gaussian distribution.
  - **Usefulness for Anomaly Detection :** Flags anomalies as data points with a very low probability of occurring under the learned multivariate Gaussian distribution, suitable for correlated features.

11.1.2 Non-Parametric

- **Inter Quartile Range (IQR)**
  - **Description :** A measure of statistical dispersion, calculated as the difference between the 75th and 25th percentiles.
  - **Usefulness for Anomaly Detection :** Defines anomalies as observations that fall outside a specified range (e.g., 1.5 times the IQR) from the first or third quartile.
- **Percentile Method**
  - **Description :** Ranks observations based on their values and defines thresholds at specific percentile points.
  - **Usefulness for Anomaly Detection :** Identifies anomalies as data points that fall into the lowest or highest specified percentiles of the data distribution.

11.1.3 Proximity-Based

- **k-Nearest Neighbors**
  - **Description :** A non-parametric method that classifies a data point based on the majority class of its 'k' nearest neighbors.
  - **Usefulness for Anomaly Detection :** Detects anomalies by identifying data points that have a low density of neighbors or are far from their 'k' nearest neighbors.
- **Local Outlier Factor**
  - **Description :** Measures the local deviation of a data point with respect to its neighbors.
  - **Usefulness for Anomaly Detection :** Identifies anomalies as data points that are significantly less dense than their neighbors, making it effective for detecting outliers in varying density regions.

11.2. Machine Learning Methods

11.2.1 Unsupervised

- **K-Means Clustering**
  - **Description :** Partitions 'n' observations into 'k' clusters where each observation belongs to the cluster with the nearest mean.
  - **Usefulness for Anomaly Detection :** Anomalies are often identified as data points that are far from any cluster centroid or belong to very small clusters.
- **Isolation Forest**
  - **Description :** An ensemble method that "isolates" anomalies by randomly building decision trees and measuring the path length required to isolate a point.
  - **Usefulness for Anomaly Detection :** Effectively detects anomalies as observations that require fewer splits to be isolated in the ensemble of trees.
- **One-Class Support Vector Machine**
  - **Description :** Learns a decision boundary that encapsulates the majority of the data points, assuming all data points are from one class.
  - **Usefulness for Anomaly Detection :** Labels data points outside the learned boundary as anomalies, particularly useful when only normal data is available for training.

11.2.2 Supervised

- **Logistic Regression**
  - **Description :** A statistical model that uses a logistic function to model a binary dependent variable.
  - **Usefulness for Anomaly Detection :** Can be used for supervised anomaly detection by classifying instances as 'normal' or 'anomalous' based on learned feature relationships.
- **Support Vector Machine**
  - **Description :** A supervised learning model that finds the optimal hyperplane to separate data points into different classes.
  - **Usefulness for Anomaly Detection :** In a supervised setting, it builds a classification boundary between normal and anomalous data, classifying new points accordingly.
- **Random Forest Classifier**
  - **Description :** An ensemble learning method that constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees.
  - **Usefulness for Anomaly Detection :** Used in supervised anomaly detection to classify instances as normal or anomalous by leveraging an ensemble of decision trees to capture complex relationships.
- **XGBoost**
  - **Description :** An optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable.
  - **Usefulness for Anomaly Detection :** A powerful supervised model that can be trained to classify data points as normal or anomalous, often achieving high performance due to its boosting capabilities.

11.3. Deep Learning Methods

11.3.1 Autoencoder

- **Description :** A type of artificial neural network used for learning efficient data codings in an unsupervised manner, consisting of an encoder and a decoder.
- **Usefulness for Anomaly Detection :** Anomalies are detected as data points that have high reconstruction errors after being passed through the autoencoder, as the model primarily learns to reconstruct normal data.

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Parametric	Z-Score	**0.9022**	0.7541	**0.0978**	0.2459	0.715	0.501	0.246	0.33
	Multivariate Gaussian	0.787	0.4821	0.213	0.5179	0.695	0.467	0.491	0.479
Non-parametric	Inter Quartile Range	0.6429	0.4718	0.3571	0.5282	0.61	0.371	0.528	0.436
	Percentile	0.7391	**0.2319**	0.2609	**0.7681**	**0.747**	**0.54**	**0.768**	**0.634**
Proximity-Based	k-Nearest Neighbors	0.7212	0.6468	0.2788	0.3532	0.616	0.336	0.353	0.344
	Local Outlier Factor	0.7857	0.4031	0.2143	0.5969	0.732	0.527	0.597	0.56

Table 5: Statistical Modelling Evaluation Table

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11.3.2 Variational Autoencoder

- **Description :** A generative model that learns a probabilistic mapping from input data to a latent space and then reconstructs the data.
- **Usefulness for Anomaly Detection :** Identifies anomalies by their low likelihood (or high reconstruction error) under the learned generative model, often providing a more robust measure of "normalness" than a standard autoencoder.

12. Evaluation

12.1. Statistical Methods

12.1.1 Evaluation Summary

- The Percentile method stands out as the best-performing model by a significant margin, 30 demonstrating superior accuracy, precision, recall, and F1-score due to its excellent 32 balance of true positives and true negatives, and low false positive rates.
- The Local Outlier Factor (LOF) is the second-best model, showing good overall performance.
- The Z-Score method, while excellent at detecting true positives, suffers severely from high false positives, making it unreliable.
- The Inter Quartile Range (IQR) and k-Nearest Neighbors (k-NN) methods are the weakest performers, showing poor results across most evaluation metrics.

⇒ Refer "Table 5" for the evaluation report.

12.2. Machine Learning Methods

12.2.1 Evaluation Summary

- Supervised models significantly outperform unsupervised models in this comparison, especially in terms of overall accuracy and balanced F1-Scores.

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.0336	**0.0195**	0.9664	**0.9805**	0.304	0.288	**0.98**	0.446
	Isolation Forest	0.7558	0.5706	0.2442	0.4294	0.663	0.413	0.429	0.421
	One-Class SVM	0.9943	0.9827	0.0057	0.0173	0.715	0.549	0.017	0.034
Supervised	Logistic Regression	0.9867	0.3867	0.0133	0.6189	0.882	0.949	0.619	0.749
	Support Vector Machine	**0.9977**	0.4751	**0.0023**	0.5249	0.863	**0.989**	0.525	0.686
	Random Forest Classifier	0.9932	0.1149	0.0068	0.8851	0.962	0.981	0.885	0.931
	XGBoost	0.9907	0.0812	0.0093	0.9188	**0.97**	0.975	0.919	**0.946**

Table 6: Machine Learning Modelling Evaluation Table

Models	Std. Threshold	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Autoencoder	0.5	0.9715	**0.9734**	0.0285	**0.0266**	0.702	0.272	**0.027**	**0.049**
	1	0.9941	0.9944	0.0059	0.0056	0.712	0.277	0.006	0.011
	2	0.9962	0.9962	0.0038	0.0038	**0.713**	0.287	0.004	0.008
	3	0.997	0.9969	0.003	0.0031	**0.713**	**0.297**	0.003	0.006
Variational Autoencoder	0.5	0.9783	0.9797	0.0217	0.0203	0.705	0.272	0.02	0.038
	1	0.9974	0.9974	0.0026	0.0026	**0.713**	0.285	0.003	0.005
	2	0.9976	0.9979	0.0024	0.0021	**0.713**	0.254	0.002	0.004
	3	**0.9977**	0.9984	**0.0023**	0.0016	**0.713**	0.219	0.002	0.003

Table 7: Autoencoder & Variational Autoencoder Evaluation Table

- XGBoost is the best-performing model overall, exhibiting exceptional balance and high scores across all key metrics (Accuracy, Precision, Recall, F1-Score).
- Random Forest Classifier is a very close second to XGBoost, also demonstrating excellent and well-balanced performance.
- Logistic Regression is a strong contender within supervised learning, offering robust performance.

⇒ Refer "Table 6" for the evaluation report.

12.3. Deep Learning Methods

12.3.1 Evaluation Summary

- Both AE and VAE models show extremely high True Positive (TP) rates but are severely hampered by very high False Positive (FP) rates across all tested thresholds.
- Consistently low Precision and extremely low F1-Scores for both models, indicates a poor balance between detecting true anomalies and minimizing false alarms.
- The Recall values are also extremely low, which, when combined with high True Positive (TP), strongly suggests an underlying issue in detecting such anomalies.
- Contrary to expectation, VAE did not perform better than AE; the latter achieved a slightly higher F1-Score and marginally better precision.

⇒ Refer “Table 7” for the evaluation report.

13. Future Work

The future work will primarily focus on the completion of advanced modeling and evaluation phases, followed by comprehensive reporting and final submission. Specifically, completing the ongoing Generative Modelling and its subsequent evaluation. Following which, the project will move into synthetic data generation and its evaluation. The generated synthetic data will then be augmented with original dataset, utilizing it for generalizing the ML models already evaluated in this report. Finally, the preparation of a Comprehensive Evaluation Report, will be based on all the models experimented upon in this report and in the future scope of the work.

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15. List of Publications

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